Sales Analysis

Tools Used: Python, Pandas, SQL, Power BI, Microsoft Excel

- **Project Introduction**

This project focuses on analysing sales data to understand business performance, identify sales trends, evaluate product profitability, and discover opportunities for improvement. The analysis involves data cleaning, exploratory data analysis (EDA), data visualization, and the development of an interactive Power BI dashboard.

Python and Pandas are used to clean and analyse the dataset, while SQL can be used to query and summarize the data. Power BI is used to present key performance indicators (KPIs), sales trends, and profitability insights through interactive visualizations.

- **Project Objectives**

The main objectives of this project are:

- To clean and prepare the dataset for accurate analysis.
- To analyse sales, costs, and profit across different products, countries, and business segments.
- To identify the best-performing products and business segments.
- To investigate unusual profit and loss values.
- To visualize sales trends over time.
- To develop an interactive Power BI dashboard that supports data-driven business decisions.

- **Dataset Description**

The dataset contains sales-related information, including business segments, countries, products, units sold, manufacturing prices, sale prices, gross sales, discounts, sales, cost of goods sold (COGS), profit, and date-related columns.

These fields help evaluate business performance from different perspectives, including revenue generation, product performance, geographical performance, and profitability.

- **Project Workflow**

The project follows these steps:

1. Data collection and initial inspection

2. Data cleaning and preprocessing
3. Exploratory Data Analysis (EDA)
4. Identification and investigation of unusual values
5. Sales and profitability analysis
6. Data visualization and dashboard development
7. Business insights and recommendations

- **Expected Outcome**

The project aims to transform raw sales data into meaningful business insights by identifying sales patterns, evaluating profitability, and highlighting areas that may require further investigation.

- **Data Cleaning and Preprocessing**

1. Introduction

Data cleaning is an essential step in the data analysis process because the quality of the data directly affects the reliability of the results. Before performing exploratory data analysis, the dataset was inspected to identify missing values, duplicate records, incorrect data types, and unusual numerical values.

Python and Pandas were used to inspect and prepare the dataset for further analysis.

2. Data Inspection

The initial inspection focused on understanding the dataset's structure, column names, data types, missing values, and numerical distributions.

The following checks were performed:

- Examining the number of rows and columns.
- Checking for missing or null values.
- Identifying duplicate records.
- Verifying the data types of numerical and date columns.
- Reviewing summary statistics to identify unusual values in sales, costs, and profit.

3. Handling Unusual Values

During exploratory data analysis, two records were identified for further investigation because their profit values were considerably different from other observations.

Case 1: Significant Loss

- Country: Canada
- Segment: Enterprise
- Product: VTT
- Date: November 2013
- Sales: 313,862.50
- Cost of Goods Sold (COGS): 354,480.00
- Profit: -40,617.50

This record shows a negative profit, indicating that the cost of goods sold exceeded sales by 40,617.50. Although the value appeared unusual, the negative profit alone was not sufficient evidence to classify it as a data error.

Case 2: High Profit

- Country: United States of America
- Segment: Government
- Product: Paseo
- Date: July 2014
- Sales: 1,159,200.00
- Cost of Goods Sold (COGS): 897,000.00
- Profit: 262,200.00

This record shows a high positive profit because sales exceeded the cost of goods sold by 262,200.00. The record was investigated rather than automatically removed.

4. Decision on Outliers

The two records were reviewed in the context of their sales, costs, units sold, discounts, and profit values.

The negative profit represented a potential business loss, while the high positive profit represented a potentially strong financial outcome. These values should not be removed solely because they are unusually high or low.

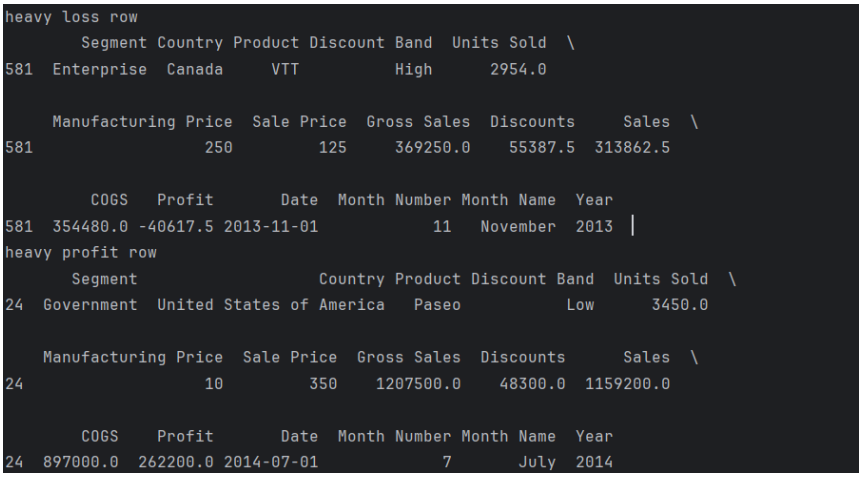
After checking the calculations and relevant business context, unusual values that are valid observations should be retained for analysis. If a record contains an actual data-entry or calculation error, it should be corrected using reliable information.

5. Conclusion

The data-cleaning process emphasizes data accuracy and preservation of meaningful business information. Investigating unusual values instead of automatically deleting them helps ensure that genuine losses and profitable transactions remain part of the analysis.

The prepared dataset can then be used for exploratory data analysis, visualization, and the development of the Power BI dashboard.

6. Insight



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heavy loss row
       Segment Country Product Discount Band  Units Sold  \
581  Enterprise  Canada     VTT          High      2954.0

     Manufacturing Price  Sale Price  Gross Sales  Discounts     Sales  \
581                  250         125     369250.0    55387.5  313862.5

         COGS   Profit       Date  Month Number Month Name  Year
581  354480.0 -40617.5 2013-11-01            11   November  2013
heavy profit row
       Segment                  Country Product Discount Band  Units Sold  \
24  Government  United States of America   Paseo           Low      3450.0

    Manufacturing Price  Sale Price  Gross Sales  Discounts      Sales  \
24                   10         350    1207500.0    48300.0  1159200.0

        COGS    Profit       Date  Month Number Month Name  Year
24  897000.0  262200.0 2014-07-01             7       July  2014
```

The analysis identified a transaction in the Canadian Enterprise segment with a loss of 40,617.50, alongside a highly profitable transaction in the United States Government segment generating a profit of 262,200.00. These records highlight differences in transaction profitability and demonstrate the importance of investigating unusual financial outcomes rather than automatically removing them.

Metric	Canada — VTT	USA —
Units Sold	2,954	3,450
Sales	313,862.50	1,159,200.00
COGS	354,480.00	897,000.00
Profit	-40,617.50	262,200.00
Profit Margin	-12.94%	22.62%

1. **Best-performing product category- Paseo**
2. **Months with unusually high profit are February, June, December, September.**
3. **Months with unusually low profit are October, July,March,January.**

4. **Regions that may need improvement are USA and Mexico.**
5. **Products with high sales but low profit Velo**